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CHAPTER 1

Introduction

1.1 Prologue

Automation everywhere in CS, we want to set up boilerplates to help our development.

Examples:

- Continuous integration
- Macros are everywhere
- Artificial intelligence

We delegate tasks to some tools.

expert system →

example plus frappant:

artificial intelligence: based on probability but not reproducible, not predicatable

several other examples in computer sciences:

...

1.2 Automation is everywhere

In the domain of formal proof automation is everywhere, and it is formally called elaboration ??

In formal proof, proofs are verbose and have lot of details: types, coercions, canonical structures, type-classes.

In theorem provers the user benefit of notations, implicit inference and tactics to ease his job.

Simply typed lambda calculus is an example of automation: an algo checks if ...

1.3 Contributions

In this manuscript we focus on an alternative type-class solver written in the elpi meta-language.

We also talk about a formalized tool...

1.3.1 Alternative type-class solver in Coq-Elpi

1.3.2 Static determinacy checking for Elpi

CHAPTER 2

State of the Art

At the very base of my PhD there are two softwares, the Rocq Proof assistant (formerly known as *Coq* [4]) and the *Elpi* [10].

2.1 The *Rocq* proof assistant

[4]

Theorem provers with HO logic programming languages: [12]

Mathematical components: [28]

Abella: [18]

SSReflect: [19]

Isabelle: [33]

2.1.1 The basis of the language

2.1.2 Elaboration

In fig. 2.1 (gently offered by Enrico), we have a skematal representation of the elaboration process.

2.1.3 Ad-hoc polymorphism: Type Classes

One of the most interesting fragment of elaboration is [46, 42, 48]

Cite Haskell [22], in caml it does not exists: `+. vs +`

IRIS: [26]

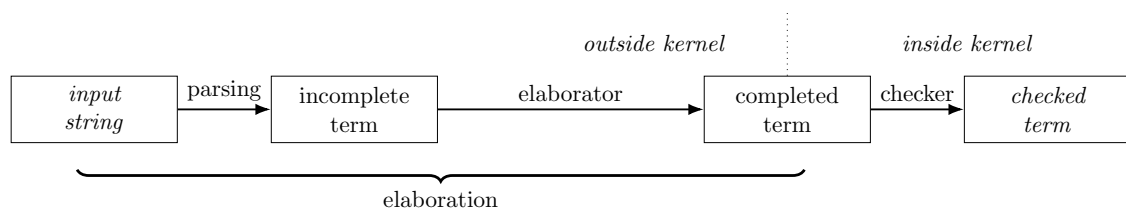


Figure 2.1: Elaboration skeme

2.2 The *Elpi* language

Elpi [10] is a dialect of λ prolog [30, 32]

Teyjus: [31]

Elf: [35]

Twelf: [36]

Lambda prolog: [39]

Mercury: [41]

2.2.1 λ -terms representation

HOAS: [34]

2.2.2 Hints about unification

Miller: [30]

Huet: [25]

2.2.3 Controlling backtracking: the ! operator

SWI prolog: [43]

Andrews: [2]

Vink: [8]

2.2.4 *CHR*: goal suspension

[17, 16]

CHR elpi: [21, section 4.3]

CHR Pfenning: [29]

2.3 *Coq-Elpi*: Rocq and Elpi interleaving

Coqelpi: [44]

Hierarchy Builder: [7]

Trackt: [5]

Equality test [20]

Derive: [45]

2.3.1 Concluding remarks

Overall, Elpi is good for elaboration

CHAPTER 3

Alternative type-classe solver in *Coq-Elpi*

Typeclasses are an elegant way to implement ad-hoc polymorphism [47], so that, the same symbol can be reused in different context while keeping the same mathematical porperties. From an implementation point of view, typeclasses are

3.1 Why typeclass resolution in *Elpi*?

[13]

Rule base automation!

3.2 From Rocq goals to *Elpi* proofs

3.2.1 Rule encoding

The hooks

3.2.2 Goal encoding

3.3 Alternative search indexing

3.3.1 Discrimination trees

Discrimination trees: Pfenning notes *

TODO: do some benchmarks

Discrimination trees: opt for lists and tree depths.

3.3.2 Some ideas on tabling

[40, 37]

*<https://www.cs.cmu.edu/~fp/courses/99-atp/lectures/lecture28.html>

3.4 Helping hihger-order unification

[14]

?: [11]

Variadic functions are typechecked in coq [6]

3.4.1 η -equivalent terms

3.4.2 β -equivalent terms

[1]

Beta problema outside the pattern can come back in LL [31]

3.5 Experience report

IRIS: [26]

CHAPTER 4

Determinacy checking for Elpi

4.1 Why determinacy?

[9, 41, 23, 24, 27]

4.2 Implementation

4.2.1 Rule checking

4.2.2 Mutual exclusion

Using discrimination tree

4.2.3 Caveats

For example, mutual exclusion is not perfect: over approximation of unification

Rule checking ignores flexible rule loading

4.2.4 Experience report

4.3 Formalization in Rocq

Using ssreflect [19]

4.3.1 The language

Input must come before outputs

Inputs must come before outputs, with Disc tree this is not mandatory, since unification is weaker.

EG:

```
pred g o:int, o:int.  
  
f 3 3 (g X X).  
f 3 4 (g 3 4).  
  
main:-  
  f X Y (g X Y).
```

In the code above, both rules can be used with the query, but, the discrimination tree, save the world

4.3.2 The tree semantics

4.3.3 The stack semantics

[\[38\]](#)

4.3.4 Semantics equivalence

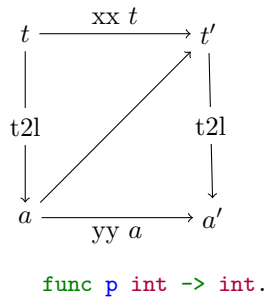
[\[3\]](#)

4.3.5 The main theorem

CHAPTER 5

Conclusion and Perspectives

[15]



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Tritre francais

Davide FISSORE